

Rtrack Data Procedure

Downloading test data from AnyMaze

- Open the .szd file in Anymaze
- Protocol > Test data report
- Check the box “The position of the animal”. Other data can be included in your test data report, but it is not necessary for Rtrack.
- Select “Show the animal’s position in pixels”

The screenshot displays the ANY-maze software interface. The top menu bar includes FILE, PROTOCOL, EXPERIMENT, TESTS, RESULTS, and DATA. Below this is a toolbar with icons for navigation (Back, Forward), clipboard (Copy, Paste), protocol management (Add item, Remove item, Load protocol, Save protocol), and view options (Hide unused items, Hide protocol list, View protocol report). The left sidebar is divided into three sections: TESTING, ADDITIONAL INFORMATION, ANALYSIS, and REPORTS. The TESTING section includes Test scheduling, Stages, Events and actions, Automatic starting of tests, What to record while testing, What to display while testing, and Test window panels. The ADDITIONAL INFORMATION section includes Treatment groups, Animal ID, Animal weights (highlighted), Treatment doses, and Fields. The ANALYSIS section includes Calculations, Analysis across time, Analysis options, and Charts. The REPORTS section includes Animal details report, Test details report, Track plots, Heat maps, Test data report (highlighted), Results reports, and Data page. The main window shows the 'Test data report' configuration. It includes a description: 'The Test data report shows things that occurred during a test in a spreadsheet format. For example, the report can list the animal's position during the test, the animal's speed during the test, each zone entry, each zone exit, etc. You can use the list below to choose what data is included in the report.' Below this is a list of data items to be included, each with a checkbox. The 'Test control' section is expanded, showing a list of items: 'Animal position and activity' (checked), 'The position of the animal' (checked), 'The position of the animal's head' (unchecked), 'The position of the animal's tail' (unchecked), 'The animal becomes hidden' (unchecked), 'The animal becomes visible' (unchecked), 'The animal's speed' (unchecked), 'The length of the animal' (unchecked), 'The relative length of the animal (relative to its mean length)' (unchecked), 'Clockwise rotations of the animal's body' (unchecked), 'Anti-clockwise rotations of the animal's body' (unchecked), 'The animal's turn angle (based on the position of its centre point)' (unchecked), and 'Orientation of the animal's head' (unchecked). Below this list are several checkboxes for 'Q1', 'Q2', 'Q3', 'Q4', 'Platform', and 'New event'. At the bottom, there are two sections for configuration: 'Choose the time format for the report:' with three radio buttons (selected: 'Show times as HH:MM:SS.sss - for example 00:10:28.153'), and 'Choose how the animal's position should be shown in the report:' with two radio buttons (selected: 'Show the animal's position in pixels').

ANY-maze - W-Maze.NARF

FILE PROTOCOL EXPERIMENT TESTS RESULTS DATA

Back Forward Copy Paste Add item Remove item Load protocol Save protocol Hide unused items Hide protocol list View protocol report

Navigation Clipboard Protocol View

TESTING

- > Test scheduling
- > Stages
- > Events and actions
- Automatic starting of tests
- What to record while testing
- > What to display while testing
- > Test window panels

ADDITIONAL INFORMATION

- Treatment groups
- Animal ID
- Animal weights
- Treatment doses
- Fields

ANALYSIS

- Calculations
- Analysis across time
- > Analysis options
- Charts

REPORTS

- Animal details report
- Test details report
- > Track plots
- > Heat maps
- Test data report
- Results reports
- Data page

Test data report

The Test data report shows things that occurred during a test in a spreadsheet format. For example, the report can list the animal's position during the test, the animal's speed during the test, each zone entry, each zone exit, etc. You can use the list below to choose what data is included in the report.

Test control

- Animal position and activity
 - ☒ The position of the animal
 - ☐ The position of the animal's head
 - ☐ The position of the animal's tail
 - ☐ The animal becomes hidden
 - ☐ The animal becomes visible
 - ☐ The animal's speed
 - ☐ The length of the animal
 - ☐ The relative length of the animal (relative to its mean length)
 - ☐ Clockwise rotations of the animal's body
 - ☐ Anti-clockwise rotations of the animal's body
 - ☐ The animal's turn angle (based on the position of its centre point)
 - ☐ Orientation of the animal's head
- Q1
- Q2
- Q3
- Q4
- Platform
- New event

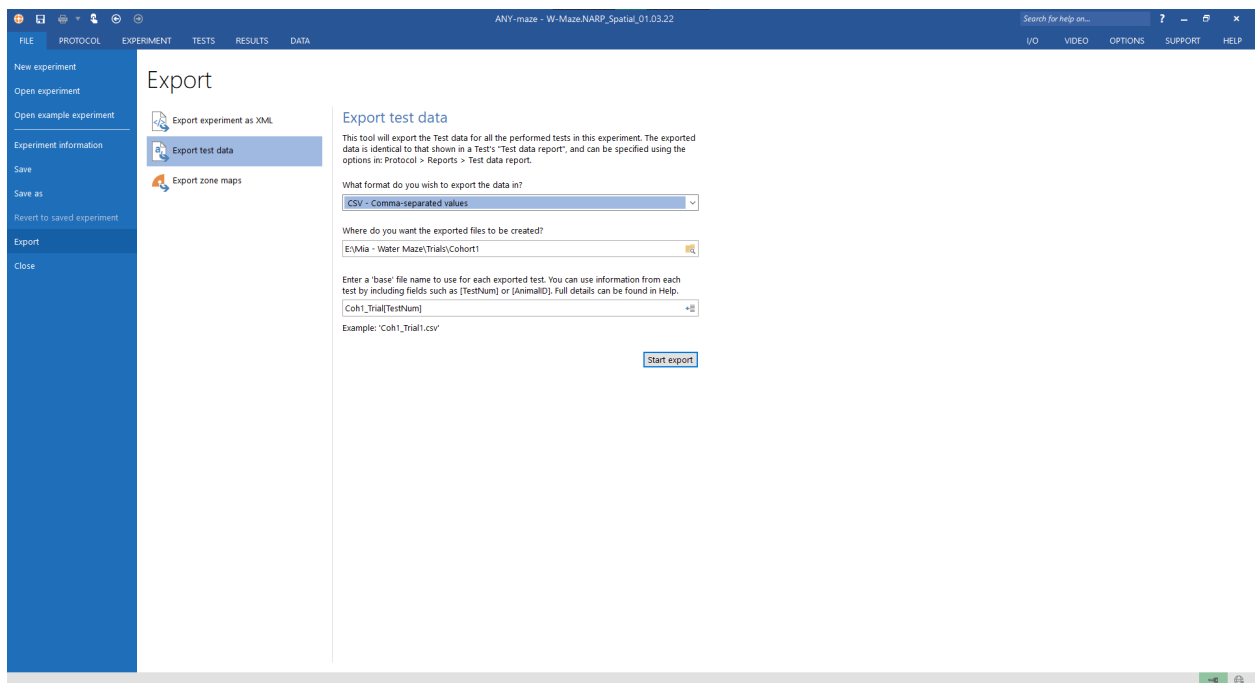
Choose the time format for the report:

- ☒ Show times as HH:MM:SS.sss - for example 00:10:28.153
- ☐ Show times as seconds only - for example 628.153
- ☐ Show times as real-time in HH:MM:SS.sss - for example 13:20:14.791

Choose how the animal's position should be shown in the report:

- ☒ Show the animal's position in pixels
- ☐ Show the animal's position in millimetres

- File > Export > Export test data
- Select the folder where you will be exporting all trials to
- Name the file following the format Coh#_Trial[TestNum]
- Click Start export



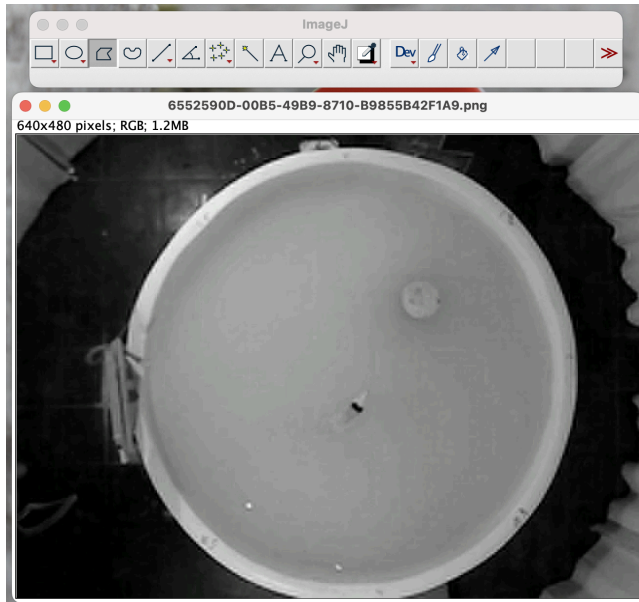
Note: Save to "All Tracks" folder

Converting videos from AnyMaze to .mp4

- Download the .szk file(s) for the video(s) that you want to convert
- In the video tab, select the files that you want to convert and select the file destination

Pool Measurements

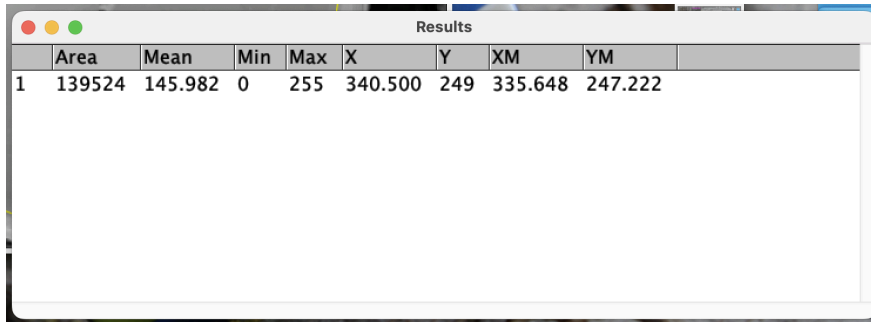
- Open the video and save a frame from it
- Open the image in ImageJ



- Use the oval measuring tool to create a circle where the inner wall of the pool is. If you are having difficulty telling where this is, it can be helpful to find a video where the rat is swimming against the inner wall to tell where this boundary is.



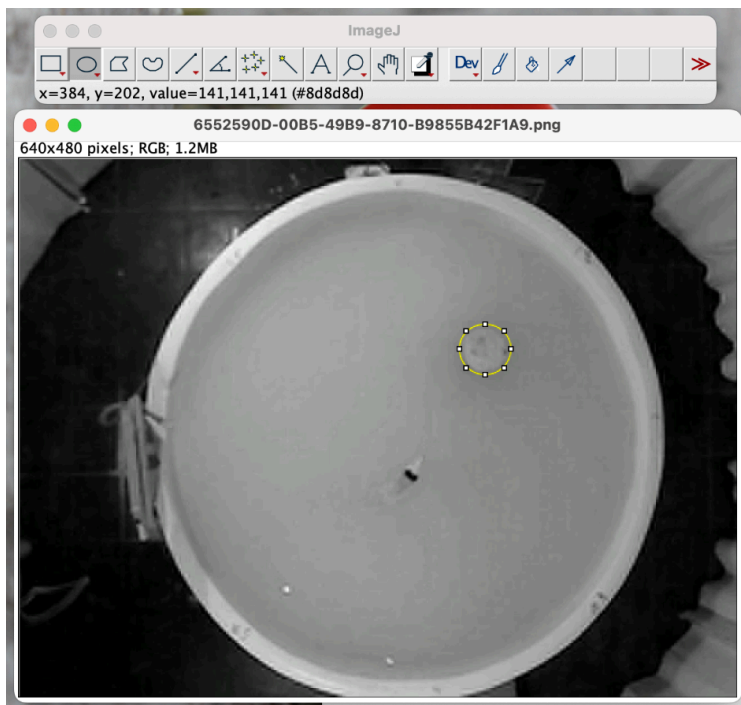
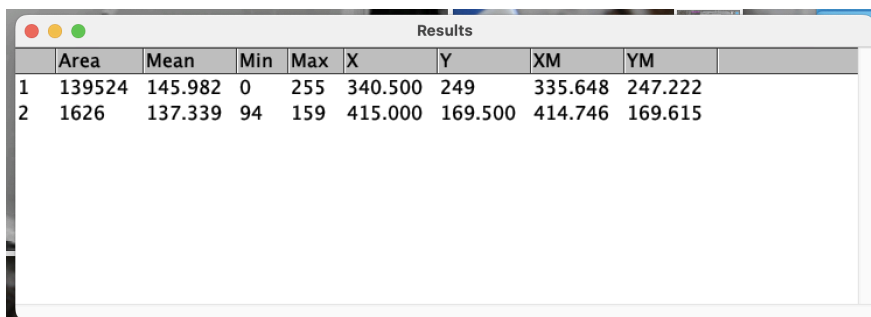
- Analyze > Measure (Command+M) opens up a window with measurements. Record the X and Y coordinates as the center of the pool, and calculate the radius of the pool using the given area.



A screenshot of the ImageJ 'Results' window. The window has a title bar with standard macOS window controls (red, yellow, green buttons). Below the title bar is a table with the following data:

	Area	Mean	Min	Max	X	Y	XM	YM
1	139524	145.982	0	255	340.500	249	335.648	247.222

- Repeat the previous two steps, but this time measure the platform.

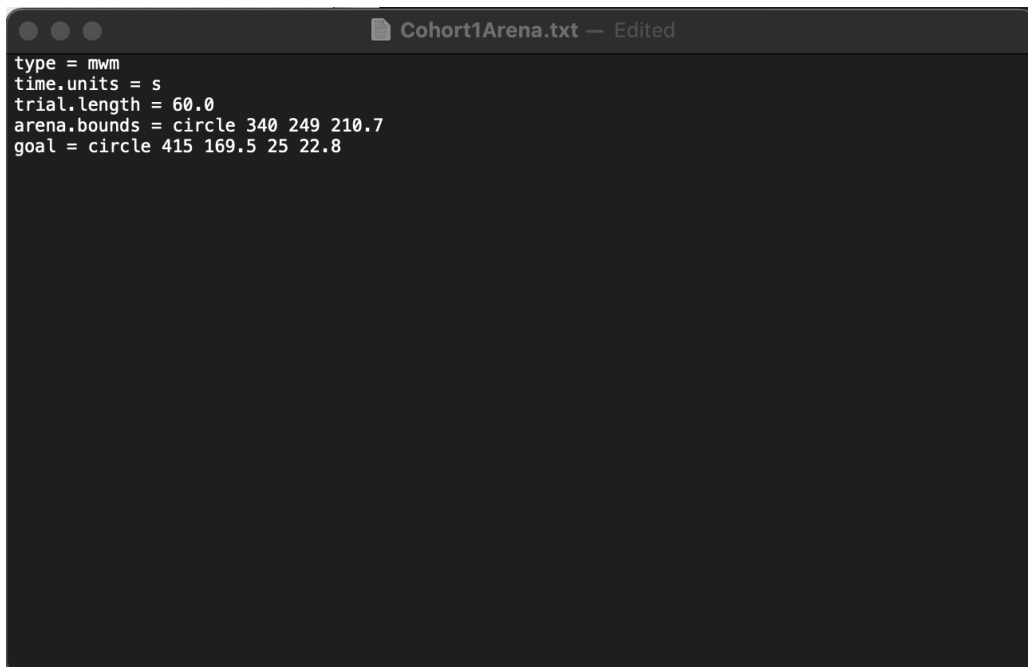



A screenshot of the ImageJ 'Results' window. The window has a title bar with standard macOS window controls (red, yellow, green buttons). Below the title bar is a table with the following data:

	Area	Mean	Min	Max	X	Y	XM	YM
1	139524	145.982	0	255	340.500	249	335.648	247.222
2	1626	137.339	94	159	415.000	169.500	414.746	169.615

Arena File

- For each cohort, create a text file titled Cohort#Arena.txt
- This file consists of 4-5 lines:
 - type: The arena type (this will be “mwm”)
 - time.units: The units that the time stamps are measured in the Trial files. This can either be a text code (“s” = seconds, “h” = hours, etc.) or a conversion factor to seconds (1 = seconds, 0.0002777778 = hours).
 - trial.length: Max amount of time that the trial can last
 - arena.bounds = The pool dimensions, which has four components separated by a space
 - The shape (only “circle” is allowed for the Morris Water Maze)
 - X coordinate of the pool center
 - Y coordinate of the pool center
 - Radius of the pool
 - goal: Platform dimensions, formatted the same as the pool dimensions
 - old.goal: Dimensions of the old platform goal, follows the same format as goal. This is only necessary to include when analyzing reversal trials.



```
type = mwm
time.units = s
trial.length = 60.0
arena.bounds = circle 340 249 210.7
goal = circle 415 169.5 25 22.8
```

Note: Save to “Arena Files” folder.

- You may have to create more than one arena file for a cohort if the pool shifted at all during the experiment, such as after it was cleaned halfway through.

Spatial Sheets

- Each cohort has an Excel file. These can usually be found in S3. They must have a Key and Spatial sheet (Probe, Cued, and Visual sheets do not matter).
- The Key sheet must have the column BarnesID or CowenID
- The Key sheet may contain other information useful for analysis, such as Age, Sex, Performance, etc.

	A	B	C	D	E	F
1	Any-Maze #	BarnesID	Age	TM	Sex	APP
2	1	10876	4mo	1 Green	F	WT
3	2	10877	4mo	2 Green	F	WT
4	3	10878	4mo	3 Green	F	APP/+
5	4	10879	4mo	4 Green	F	APP/+
6	5	10880	4mo	5 Green	F	APP/+
7	6	10881	4mo	6 Green	F	WT
8	7	10882	3mo	1 Purple	F	APP/+
9	8	10883	3mo	2 Purple	F	WT
10	9	10884	3mo	3 Purple	F	WT
11	10	10885	3mo	4 Purple	F	WT
12	11	10886	3mo	5 Purple	F	APP/+
13	12	10887	3mo	6 Purple	F	APP/+
14	13	10888	2mo	1 Red	F	WT
15	14	10889	2mo	2 Red	F	APP/+
16	15	10890	2mo	3 Red	F	APP/+
17	16	10891	2mo	4 Red	F	WT
18	17	10892	2mo	5 Red	F	WT
19	18	10893	2mo	6 Red	F	APP/+

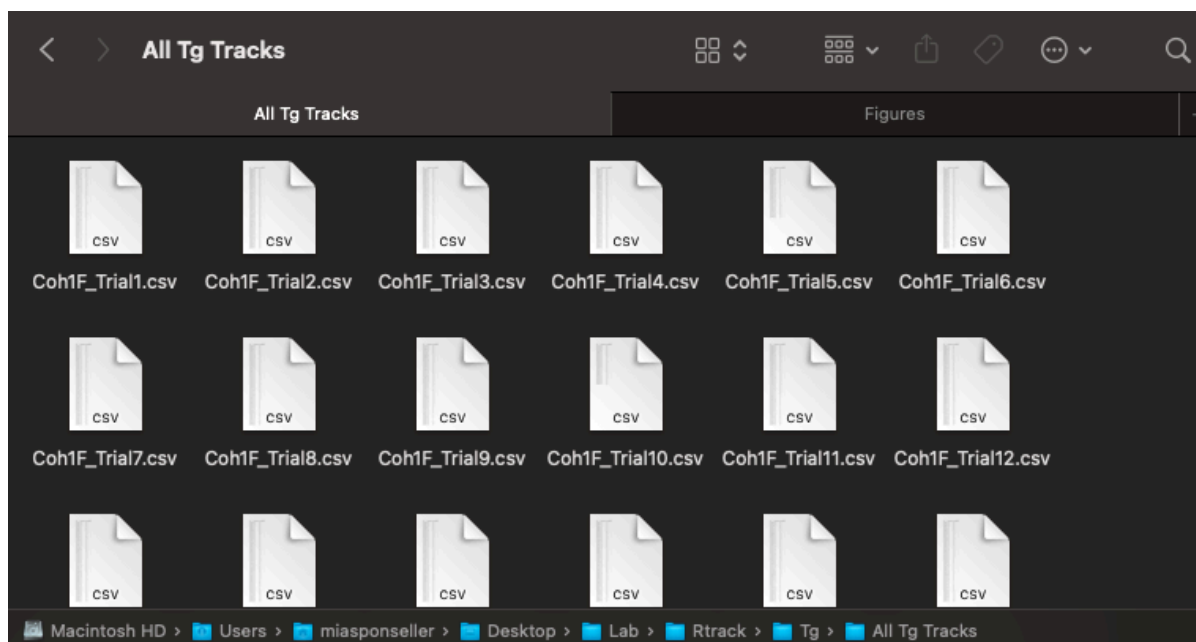
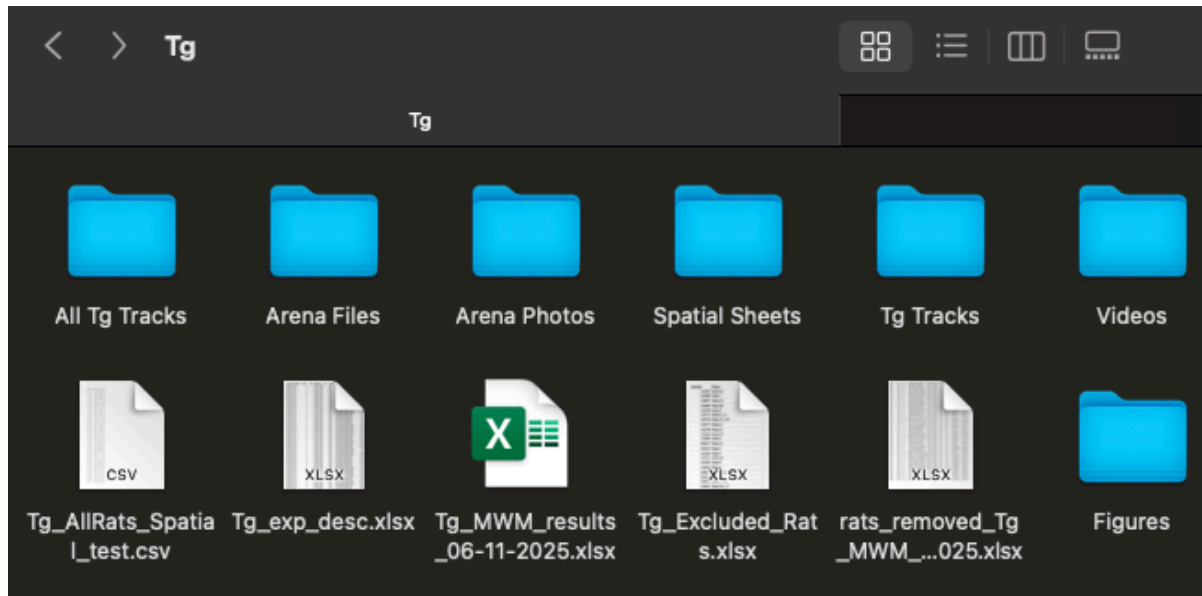
- The Spatial sheet must contain columns Test, Animal, and Trial
 - If the Test column is missing and you have to manually add it, be sure that the correct test number is assigned to each trial, as there can be instances where a test is deleted or redone and a number is skipped.
- The Spatial sheet may contain CIPL, Duration, Distance, etc.

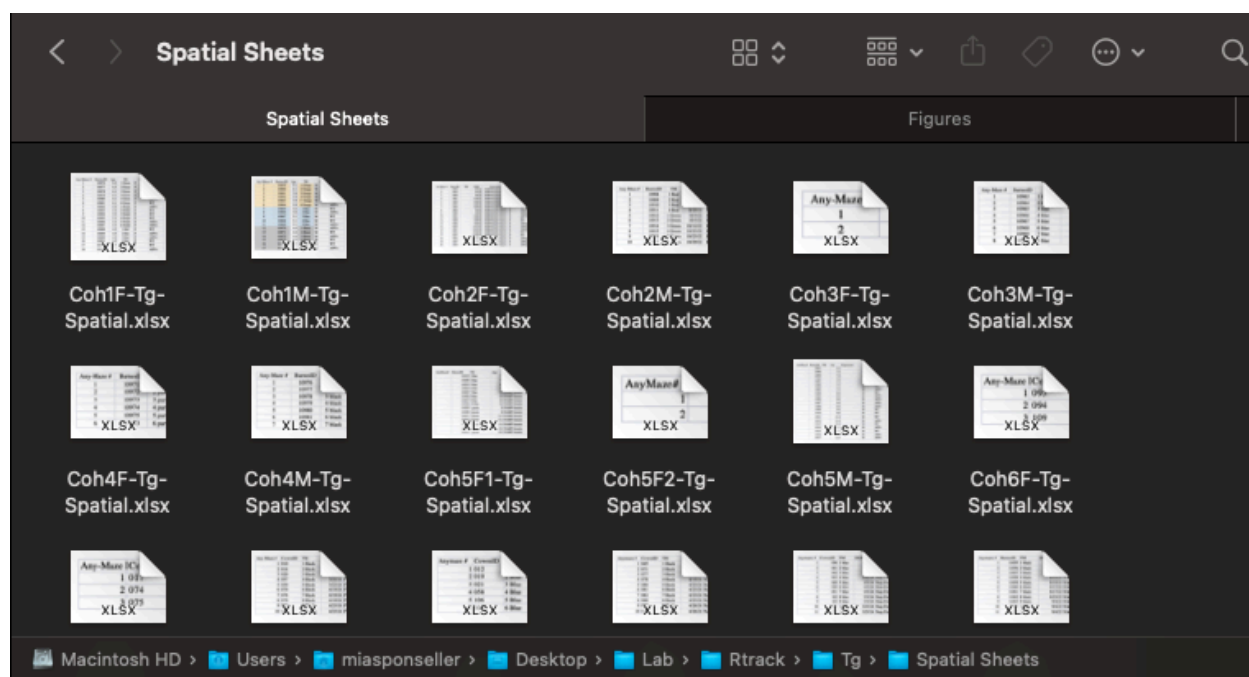
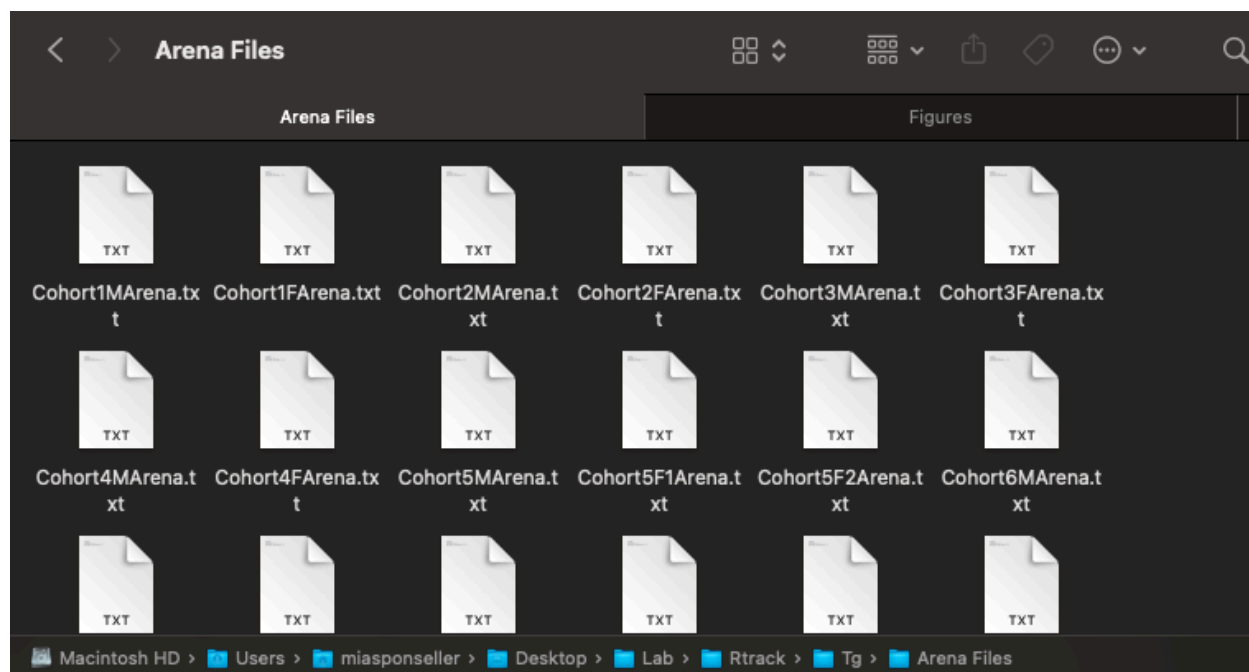
	A	B	C	D	E	F	G	H	I	J
1	Test	Animal	Stage	Trial	Apparatus	Duration	Distance	Mean speed	Path efficiency	CIPL
2	1	10876	Spatial	1	Pool	60	14.994	0.25	0.014	50.75
3	2	10876	Spatial	2	Pool	60	17.958	0.299	0.061	46.95
4	3	10877	Spatial	1	Pool	60	13.7	0.228	0.014	44.49
5	4	10877	Spatial	2	Pool	60	19.446	0.324	0.047	46.45
6	5	10878	Spatial	1	Pool	4.8	1.028	0.214	0.436	1.31
7	6	10878	Spatial	2	Pool	14.3	3.167	0.221	0.289	5.83
8	7	10879	Spatial	1	Pool	49.6	14.841	0.299	0.088	39.18
9	8	10879	Spatial	2	Pool	39.3	10.71	0.272	0.051	20.08
10	9	10880	Spatial	1	Pool	45.6	13.455	0.295	0.058	24.12
11	10	10880	Spatial	2	Pool	8.7	6.083	0.699	0.163	4.35
12	11	10881	Spatial	1	Pool	60	11.505	0.192	0.023	28.47
13	12	10881	Spatial	2	Pool	60	20.456	0.341	0.046	39.11
14	13	10882	Spatial	1	Pool	60	16.986	0.283	0.03	41.39
15	14	10882	Spatial	2	Pool	56	13.153	0.235	0.083	40.36

Note: Save all spatial sheets to "Spatial Sheets" folder. Name then as Coh#-EXPERIMENT-Spatial.xlsx

File organization

Here is how I organized my files. I have a main folder (Tg here) that I put all the folders/files associated with that experiment into. The folders that you must have are All Tracks, Arena Files, and Spatial Sheets.





AllRats_Spatial CSV File

- Once your files are organized as directed above, you can run AllRatsCSV.R
- https://github.com/SahanaSrivathsa/Barnes_WaterMaze/blob/main/RCode/AllRatsCSV.R

Experiment Description File

- This must be run after AllRatsCSV.R has been run
- https://github.com/SahanaSrivathsa/Barnes_WaterMaze/blob/main/RCode/ExperimentDescriptionFile.R

Rtrack Strategy Export

- Must be run after generating AllRats_Spatial.csv and the experiment description file
- Ideally, after the tracks have been processed, you should get a message that the file has been created successfully and no warnings. Unideally, you will get a warning(s) saying that no data could be found for a trial because it either doesn't contain path data or doesn't exist. If it does exist, this warning could appear due to there not being enough data points (ie, there are only three data points).
- https://github.com/SahanaSrivathsa/Barnes_WaterMaze/blob/main/RCode/StratExport.R

Remove Rats from Rtrack Strategy Sheet

- Use if you need to remove rats after generating the strategy sheet (such as if a rat doesn't have valid path data for a trial, and you want to remove that rat from analysis so that you're only including rats with all 24 trials)
- After running this, you will need to rerun the Experiment Description File Code and Rtrack Strategy Export code
- https://github.com/SahanaSrivathsa/Barnes_WaterMaze/blob/main/RCode/RemoveRats.R